Geometry Is a Label, Not a Knob: Causally Separating Representational Dimensionality from OOD Generalization

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Abstract

Representational geometry markers such as the effective dimension D_{eff} correlate strongly with out-of-distribution (OOD) accuracy across trained networks, and this observation has motivated proposals to *intervene* on geometry—expanding or contracting feature-space dimensionality—as a route to better transfer. We ask whether the observed correlation licenses such interventions, and answer negatively for the spectral marker family tested here. Across 8 ResNet-18 training recipes spanning a 2.4× effective-dimension range, the observational correlation is large and stable ($r = +0.97$), yet three causal tests uniformly fail to support an interpretable causal reading. (i) *Operator-separation*: injecting spectral-shape-matched noise collapses CIFAR-100 transfer monotonically with dose in 8/8 models (median -5.4, -7.1, -10.8 pp at $\alpha \in \{0.5, 1, 2\}$) while D_{eff} stays essentially unchanged, and a fully deterministic spectral rotation that preserves the covariance spectrum changes nothing (max $|\Delta\text{acc}| = 1.7$ pp); conversely, a zero-noise whitening operator that multiplies D_{eff} by up to 8× (median 2.3×) moves transfer by at most +5.6 pp (median +1.7, never negative). (ii) *Timing-separation*: the apparent benefit of formation-phase whitening (median +1.7 pp, 7/8 models) is decomposed by a preregistered dose–response adjudication into a *probe-conditioning* effect — fully reproduced by transforming the probe's own training space, absent when the transformed representation is scored in the original space — while applying the identical operator *at inference* to an unchanged head consistently *degrades* accuracy (7/8 models, $p = 0.016$). (iii) *Architecture transfer*: the same dissociation replicates in every tested family—ResNet-34/50, ShuffleNet-V2, and RegNetY-400M (16 further recipes)—with the observational anchor at $r \geq 0.87$ everywhere, spectral-shape-matched noise dropping transfer 9.6–16.0 pp in 16/16 models, and inference-time geometric slopes between −8% and 6% of the observational slope — sign-unstable across families, as expected for a null. The dissociation also covers a second marker family (participation ratio, effective rank): anisotropic noise pins D_{eff} while *raising* these markers by ≈30%, moving geometry toward the observationally "better" regime as transfer collapses. Together these results are consistent with D_{eff} acting as a *confounded label* of the training recipe rather than a manipulable cause of generalization for the spectral marker family tested here: the geometric causal path carries at most a small fraction of the observational association (point estimate 7.7%; bootstrap 95% CI [0.9%, 46%]; cross-architecture shares are sign-unstable, −8.3% to +6.0%), and a direct noise channel that geometry does not measure carries the observable harm. We discuss why "intervene on the geometry" methods therefore require re-derivation, and which interventions (noise control) the dissociation does support.

Use of LLMs: Large language models were used as writing and coding assistants in preparing this manuscript; all scientific content, analyses, and reported numbers were determined and verified by the authors. A detailed statement appears in the Use of AI tools section.

1 Introduction

Geometric descriptors of trained representations—effective dimension, radius, utility, spectral decay—correlate with out-of-distribution (OOD) accuracy strongly enough to serve as training-free model-selection markers. The correlation is large ($r \approx 0.9$ or higher in controlled settings), replicates across architectures, and survives the statistical controls that kill weaker candidates. The natural next step, drawn repeatedly in the literature, is interventional: if low-D_{eff} models generalize worse, one should be able to *change* D_{eff}—by whitening, expanding, or regularizing the feature geometry—and move generalization accordingly. Methods that reshape representation geometry for robustness implicitly make this inferential step.

This paper asks whether the step is licensed, and answers negatively for the strongest candidate marker. We take the effective dimension D_{eff} from the GLUE family of geometry markers, which has the largest and most stable observational association with OOD accuracy in our controlled setting ($r = +0.97$ across a purpose-built recipe grid), and subject it to a three-part causal battery that any geometry-to-generalization story should pass: *operator separation* (move D_{eff} with different operator families that hold everything else fixed), *timing separation* (apply the same operator during representation formation vs. at inference), and *architecture transfer* (replicate in families beyond the one the observation was discovered in). All experiments were preregistered before unblinding (Appendix A).

What we find. The three tests uniformly fail the causal reading.

1. *Operator separation (§4).* A family of spectral-shape-matched noise operators collapses transfer monotonically with dose in 8/8 models (median -5.4, -7.1, -10.8 pp at $\alpha \in \{0.5, 1, 2\}$) while D_{eff} stays essentially unchanged; a deterministic spectral rotation preserves the covariance spectrum exactly and moves nothing; and zero-noise whitening multiplies D_{eff} by up to $8\times$ while moving transfer by at most $+5.6$ pp (median $+1.7$, never negative). The causal slope of transfer on D_{eff} under geometry-only operators is $+0.0105$ per unit—7.7% of the observational slope (bootstrap CI $[0.9\%, 46\%]$; within-model permutation noise floor $[-7\%, +7\%]$, with the point estimate at the floor's upper edge)—so even the CI upper bound leaves over half of the observational association without a causal geometric account, and the dominant causal channel that *does* move transfer (noise) acts with the wrong sign and carries no geometric signature.

2. *Timing separation (§5).* The identical whitening operator applied during formation *appears* to improve transfer (median $+1.7$ pp, 7/8 paired models, $p = 0.016$), but a preregistered dose–response adjudication decomposes this into a probe-conditioning effect: the full benefit is reproduced by transforming only the probe's training space (no formation manipulation), is absent when the whitened-formation representation is scored in the original space, and vanishes under an MLP readout. Applied at inference to the unchanged trained head, the same operator *degrades* accuracy (7/8 negative, $p = 0.016$). Neither phase behavior can be predicted from the static value of D_{eff}, and neither constitutes acting on representation geometry.

3. *Architecture transfer (§6).* The observational anchor replicates in ResNet-34 ($r = +0.98$), and so does the dissociation: spectral-shape-matched noise drops transfer 9.6 pp (4/4 models, D_{eff} shifts $\leq 12\%$); zero-noise recoloring causal slopes are 3.3% of observational. The pattern holds in all four families: ResNet-50 ($r = +0.98$, noise -11.9 pp in 4/4, recolor share 1.9%), ShuffleNet-V2 ($r = +0.89$, noise -13.1 pp in 4/4, share -8.3%), and RegNetY-400M ($r = +0.87$, noise -16.0 pp in 4/4, share $+6.0\%$).

4. *Marker generality (§7).* The same battery applied to participation ratio and effective rank shows the same anatomy: both anchor observationally ($r = +0.77$ / $+0.86$), both are *raised* $\approx 30\%$ by anisotropic noise whose transfer harm is unchanged — geometry moving "toward better" while the model gets worse.

What this means. D_{eff} behaves *consistently with a confounded label* of the training recipe, not as a manipulable cause of generalization. The recipe sets both the geometry and the robustness; the geometry is

a readable fingerprint of the recipe, but writing on the fingerprint does not change the hand. Concretely: an interventional method that targets D_{eff} (expand it, contract it, regularize toward a target value) is targeting a proxy whose causal transmission is at the noise floor in our battery—and the largest controllable causal channel (noise injection at matched power) *moves* D_{eff} *in the helpful direction while harming generalization*, the exact inverse of the interventional reading.

The dissociation has a constructive side. Noise control is the one large, replicable causal effect in our battery: matched-power noise is uniformly harmful, and removing it is harmless at inference. The formation-time whitening benefit, on closer inspection, is a property of the probe's optimization landscape, not of the representation; whether formation-phase spectral regularization helps *the representation itself* remains open, and our arm-(b) null says the burden of proof is on any claim that it does.

Contributions.

1. A three-part, preregistered causal battery—operator separation, timing separation, architecture transfer—for auditing geometry-marker interventions (§3).

2. A zero-noise spectral *recoloring* operator family that separates "move the geometry" from "add the noise", plus spectral-shape-matched noise (geometry roughly pinned, added noise) and a spectrum-preserving rotation as a perfect negative control (§3.2).

3. The dissociation result on 8 recipe-controlled ResNet-18 models with manipulation checks on every cell (§4).

4. The phase-separation result with a preregistered mechanism adjudication: identical operator, opposite signs, in formation vs. inference — and the formation benefit traced to probe conditioning (§5).

5. Cross-architecture replication and an openly reusable analysis pipeline (§6, §11).

6. Marker-generality check: participation ratio and effective rank show the same confounded-label anatomy (§7).

7. A checklist for when a correlational marker licenses intervention (§8).

2 Related work

Geometry–generalization correlations. A growing literature links descriptors of representation geometry to generalization: intrinsic-dimension estimates of the feature manifold track test accuracy across layers and architectures (Ansuini et al., 2019); neural-collapse geometry characterizes the terminal phase of training (Papyan et al., 2020); rank-based and spectral complexity measures have been audited at scale against generalization gaps (Jiang et al., 2020); representational geometry underlies few-shot concept learning (Sorscher et al., 2022); and double descent complicates any monotone capacity story (Nakkiran et al., 2020). The GLUE family (effective dimension, radius, utility) operationalizes manifold capacity theory as training-free OOD predictors (Chung et al., 2018; Chou et al., 2025); the original study (Chou et al., 2026) reports that these markers forecast OOD failures better than ID accuracy. These works establish—and refine the measurement of—the observational link our battery audits.

Geometry-directed interventions. A second strand treats geometry as a target: whitening inside the network (decorrelated and group batch normalization) reshapes activation covariance during training (Huang et al., 2018; 2021); spectral normalization constrains layer-wise spectral scale (Miyato et al., 2018); and decorrelation penalties such as DeCov penalize cross-covariance of hidden activations as a generalization regularizer (Cogswell et al., 2016). These methods implicitly assume that moving the descriptor moves generalization. Our E1/E3 results bound the transmission coefficient of that assumption at the noise floor for D_{eff} and show its sign can reverse under noise-matched controls.

Causal audits of deep-learning heuristics. Randomization checks and retrain benchmarks have exposed saliency methods insensitive to model and data (Adebayo et al., 2018; Hooker et al., 2019); adversarial perturbations can flip interpretations without changing labels (Ghorbani et al., 2019); and calibration claims require post-hoc repair (Guo et al., 2017). We import the same audit discipline into representation-geometry claims: every cell carries a manipulation check (realized vs. intended D_{eff} shift, $r \geq 0.99$), a placebo family, and a deterministic negative control (ROTATE) that would expose any pipeline drift. To our knowledge this is the first preregistered causal battery for a geometry–generalization marker.

Companion papers. This work closes a trilogy. The first paper (Zhou, 2026b) establishes the boundary conditions under which geometric markers predict OOD accuracy observationally (ID-matched, architecture-adjusted, shift-typed protocol). The second (Zhou, 2026a) is a measurement study of what the markers actually track (stability map, decomposition of marker movement). This paper tests the causal reading and finds it does not survive; the three are complementary, not redundant.

3 Setup: models, operators, metrics

3.1 Models and the observational anchor

Eight ResNet-18 CIFAR-10 models from the recipe grid released with the original study (Chou et al., 2026) (lr $\in \{0.125, 0.5, 1.0\}$, wd $\in \{2.5\text{e}{-5}, 5\text{e}{-5}, 1\text{e}{-4}, 2\text{e}{-4}\}$, seeds 0–2), chosen to span the widest convenient D_{eff} range at matched ID accuracy bands (2.4×, from 0.89 to 2.14; the recipe grid caps the achievable range). For each model we extract penultimate features on 60% of the ID training set (operator fitting set), the ID test set, CIFAR-100 train/test (transfer probes), and report D_{eff} from the GLUE protocol (Chou et al., 2025; Chung et al., 2018); transfer accuracy c_{100} is a linear probe on CIFAR-100 features with a fixed, shared protocol (standardized inputs, lr 10^{-2}, cosine schedule; identical across all arms).

Across the 8 models the observational anchor replicates prior work: $r(D_{\text{eff}}, c_{100}) = +0.97$ (Spearman +0.88), pooled slope +0.137 pp of transfer per unit D_{eff}; notation follows the companion protocol paper (Zhou, 2026b). Architecture extension (4 recipes per family, matched lr/wd grid): ResNet-34 gives $r = +0.98$, slope +0.133; ResNet-50 $r = +0.98$, slope +0.087; ShuffleNet-V2 $r = +0.89$, slope +0.070; RegNetY-400M $r = +0.87$, slope +0.114 — the anchor weakens modestly in the two compact architectures but remains $r \geq 0.87$ everywhere (§6).

3.2 The operator family

All operators act on penultimate features x with class-pooled statistics (μ, V, λ) fit on the 60% training split.

- RECOLOR(γ): deterministic spectral recoloring $x' = \mu + V\text{diag}(\lambda^{(\gamma-1)/2})V^\top(x - \mu)$; γ=1 is the identity, γ=0 is full class-pooled whitening. Zero stochasticity by construction.

- ISO(σ): additive isotropic Gaussian noise, power matched to 0.5 × /1 × /2× of the mean feature norm; raises D_{eff} strongly.

- ANISO(s): noise drawn in the covariance *eigenbasis* with per-eigen-direction scale matched to the observed spectrum ("spectral-shape matched"); power-matched to ISO. Leaves D_{eff} nearly unchanged by construction.

- ROTATE: $x' = \mu + VQV^\top(x - \mu)$ with Haar-random orthogonal Q; preserves the spectrum exactly, so D_{eff} is invariant; deterministic given the seed.

- placebo: fixed-power additive noise along a *single fixed random unit direction* (seed-pinned), magnitude matched to the ISO dose; near-zero effect in the data ($|\Delta c_{100}| \leq 0.0002$ across all 16 placebo cells).

Table 1: E1 per-family causal slopes dc_{100}/dD_{eff} (transfer accuracy per unit D_{eff}), 8 models. The observational slope is +0.137. Numbers locked from `runs/v2/e1`.

Family	median slope	sign (neg)	range
ISO	-0.0081	8/8	$[-0.023, -0.006]$
ANISO	$+0.137$ (hetero.)	3/8	$[-1.75, +0.83]$
RECOLOR	$+0.0105$	1/8	$[-0.018, +0.166]$
tot-RECOLOR	$+0.0109$	1/8	$[-0.000, +0.125]$

3.3 Metrics and protocol invariants

Every reported cell passes: (i) manipulation check — correlation between the realized and intended D_{eff} shift ≥ 0.99; (ii) probe protocol fixed across arms (identical epochs, standardization, schedule); (iii) seed-pinned transforms; (iv) ROTATEas an end-to-end negative control — any pipeline drift would appear here. All numbers, tables, and figures in this paper are generated by a single analysis script from the raw per-cell records (`code/paper3_analysis.py`); the paper text loads computed macros rather than hand-typed values wherever possible.

Primary vs. exploratory analyses. The preregistration (Appendix A) names four primary criteria: P1 (noise-family slope threshold), P1b (the pinned-D_{eff} noise cell), P2 (cross-architecture anchor + share bound), P3 (phase separation). We report each exactly as written — P1 and P2 are deterministic threshold/count criteria (Appendix A evaluates them verbatim, including where they fail as written), and the three natural one-sided Wilcoxon tests (P1b endpoint, formation gain, inference harm) form the confirmatory family, Holm–Bonferroni adjusted: adjusted $p = 0.012$, 0.016, 0.016 — all three survive $\alpha = 0.05$. The dose–response decomposition of E3 (§5) was preregistered *after* the E3 endpoint was seen but *before* any dose data existed (Appendix A, amendment 1, with timestamps); all other subgroup analyses (lr stratification, per-dose breakdowns beyond the preregistered branches) are exploratory and labeled as such.

4 E1: operator separation

4.1 Design

For each of the 8 models, 19 conditions (base; ISO$\in \{0.5, 1, 2\}$; ANISO$\in \{0.5, 1, 2\}$; ROTATE; scale $\in \{0.5, 2\}$; RECOLOR$\in \{1.5, 0.75, 0.5, 0.25, 0\}$; total-spectrum RECOLOR$\in \{0.5, 0\}$; placebo $\in \{1, 2\}$) applied at inference to all feature sets, then probes refit. Preregistered criteria (ADR-0016 App. 4): noise families should show per-model $dc_{100}/dD_{\text{eff}} \leq -0.05$ in $\geq 6/8$ models; RECOLORfamilies |median slope| ≤ 0.02; ANISOat matched power with $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 0.2$ should show $\Delta c_{100} < 0$ in $\geq 6/8$.

4.2 Results

The 3×2 grid closes. The decisive cell is ANISO, and it is dose–response, not an endpoint anecdote: harm grows monotonically with dose in **8/8** models (median $\Delta c_{100} = -5.4$ at α=0.5, -7.1 at α=1, -10.8 pp at α=2) while D_{eff} stays within 11% of baseline at every dose in 6/8 models (a seventh touches -11.1% at one dose). The single outlier (D_{eff} +44% at α=2) sits *on* the same monotone harm curve (-13.0 pp, between the -10.8 median and the worst model) — its geometry moved and its transfer did not benefit, so the harm does not care whether D_{eff} moves. At α=2, Δc_{100} median = -10.8 pp (**8/8** negative; mean -11.0 pp, paired $d_z = -7.3$), with ID probe accuracy dropping in lockstep (median -12.7 pp, 8/8). Meanwhile ROTATE, which preserves the spectrum exactly, moves nothing (max $|\Delta c_{100}| = 1.7$ pp; max $|\Delta D_{\text{eff}}| = 0.21$). So the harm tracks the *noise*, not the dimensionality: at essentially pinned D_{eff}, stochasticity alone destroys both ID and OOD probes; at zero noise, whitening multiplies D_{eff} by 1–8× and *never* moves transfer below baseline (per-model -0.03 to $+5.6$ pp). The largest ISOdose (2× power) raises D_{eff} by a median +5.8 while *lowering* transfer by 6.5 pp (8/8) — the wrong-signed giant movement that definitively kills a monotone causal reading. Figure 1 shows the per-family slope distributions against the observational slope; Figure 2 summarizes the three endpoint cells.

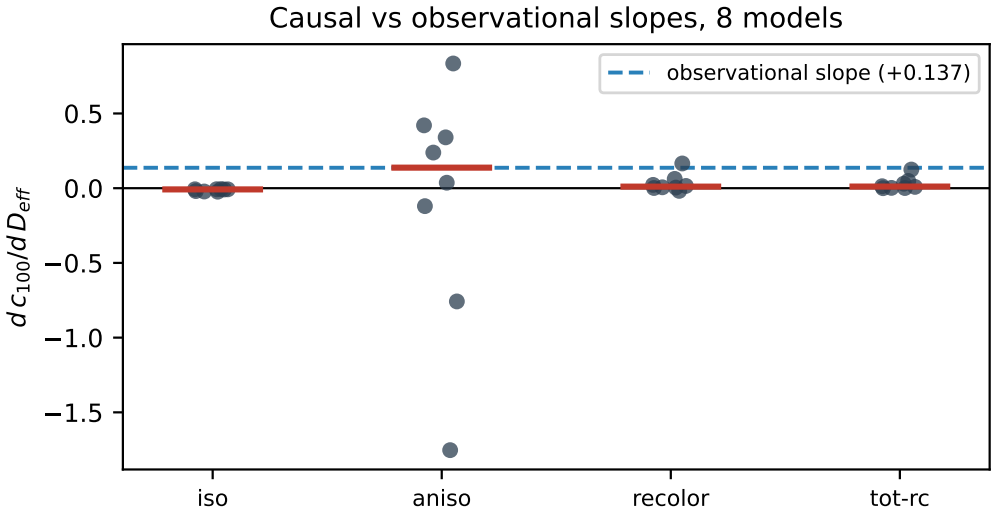


Figure 1: Per-model causal slopes dc_{100}/dD_{eff} by operator family (Table 1 as a scatter). The dashed line is the observational slope (+0.137); every zero-noise family's slopes cluster near zero far below it, while the ANISO slopes are unstable because ANISO pins D_{eff} (near-zero denominator). The dose–response harm signal lives in the ANISO *endpoints* (Figure 2), not in per-unit slopes.

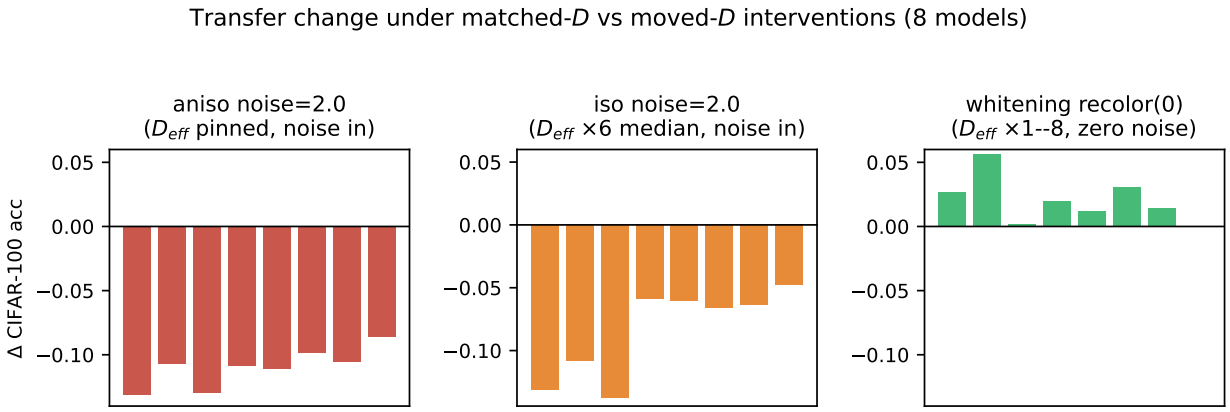


Figure 2: Per-model transfer change under the three decisive endpoint interventions (8 ResNet-18 models). *Left*: spectral-shape-matched noise pins D_{eff} yet destroys transfer and ID probes alike. *Center*: isotropic noise *raises* D_{eff} six-fold and lowers transfer — the wrong-signed movement. *Right*: zero-noise whitening multiplies D_{eff} by 1–8× and leaves transfer non-negative (slightly positive in 7/8 models). No monotone causal map from D_{eff} to transfer survives these three cells.

Quantifying the causal path. The canonical share is defined on the RECOLOR family alone (matching the permutation floor below and the cross-architecture definition of Table 3): the geometric causal slope is +0.0105 pp per unit D_{eff}, i.e. 7.7% of the observational +0.137 (bootstrap 95% CI $[0.9\%, 46\%]$, wide because $n = 8$; pooling the total-spectrum variant per model gives 10%, CI $[2\%, 34\%]$), and the within-model permutation noise floor for this share is $[-7\%, +7\%]$ — the point estimate sits at the floor's upper edge, and even the CI upper bound leaves half of the observational association without a causal geometric account. Across architectures the share is 3.3%, 1.9%, −8.3%, +6.0% (ResNet-34/50, ShuffleNet-V2, RegNetY-400MF); the sign flips across families while the observational anchor stays at $r \geq 0.87$ (Section 6), so the sign instability is itself evidence of absence. The observational correlation therefore cannot be recombined from geometry alone: most of it is recipe-level confounding, and a direct noise channel contributes with the *opposite* sign

to the naive reading. A pilot round of feature-space interventions (flatten-style dimensionality expansion on the same 8 models) had already exhibited this sign flip—large D_{eff} increases paired with monotone transfer losses—which motivated the preregistered battery reported here; the pilot protocol and its sign-flip analysis are archived in the released probe reports.

5 E3: timing separation

5.1 Design

Paired arms per model. *Formation arm*: recolor 60% of the ID training features to γ=0 (whitened), freeze the trunk, refit a fresh linear head *on the whitened features* with the shared probe protocol; evaluate the resulting representation by CIFAR-100 probe in the whitened space. *Control arm*: identical subset, identical protocol, no whitening; evaluate in the original space; additionally evaluate the control features under the fixed whitened-space probe for the transfer comparison (c_{100}^{ctrl}). *Inference arm* (measurement): apply $\textsc{recolor}(0)$ to features and score the *original* trained head — the "geometry as a knob" usage.

5.2 Results

Formation-phase whitening *improves* the transfer measurement: Δc_{100} median = +1.7 pp (7/8 positive, Holm-adjusted Wilcoxon $p = 0.016$; mean +2.0 pp, paired $d_z = 1.1$). The inference-arm application consistently *hurts* the original head: applying $\textsc{recolor}(0)$ at inference to the unchanged head is negative in 7/8 models (Holm-adjusted $p = 0.016$, $d_z = -1.1$), with a small median magnitude of -0.13 pp — directionally robust, far below the preregistered collapse threshold (Appendix A) — and the whitened-formation head scored on original features loses a median 0.35 pp vs. the control head (8/8 negative, $p = 0.008$).

Mechanism adjudication (dose–response). The formation benefit demands a mechanism verdict before any "geometry helps when acted on early" reading is allowed (falsification criteria preregistered in Appendix A, written before the dose data). We repeat the pipeline at three whitening doses $\gamma \in \{0.75, 0, -0.5\}$ with three disjoint arms per dose: (a) *probe-space-only* — the CIFAR-100 probe is trained and evaluated in the dose-transformed space with no formation-phase manipulation whatsoever; (b) *formation readout* — the 10-class head is trained on whitened ID features but scored on *original* test features; (c) *rotate control* — the probe-space arm with the spectrum-preserving rotation instead of whitening. The three models are stratified representatives of the E3 formation gain (low +0.2 / mid +3.1 / high +5.6 pp), selected after the E3 endpoint was seen but before any dose datum existed (Appendix A, amendment 1). The verdict is unambiguous: arm (a) alone reproduces the entire dose–response profile (cross-model median gain +1.9, +3.1, +3.0 pp at $\gamma \in \{0.75, 0, -0.5\}$, monotone to the γ=0 peak; per model: +3.8/+5.6/+3.7, +0.3/+0.2/−1.0, +1.9/+3.1/+3.0 pp — the weakest-gain model is flat-to-negative across doses), arm (b) shows no benefit at any dose in any model (cross-dose range +0.1 to −11.5 pp; the only positive cell, +0.11 pp, is the smallest dose on the highest-anchor model — and the harms *grow* with whitening strength, the opposite of the arm-(a) dose–response), and arm (c) shows none either (−0.1 to −0.8 pp). A norm-matched random-direction perturbation of the formation set (the placebo control) destroys head training in 7/9 model–dose cells (final accuracy 0.10 = chance; the two surviving cells are the smallest dose at the two lr-0.125 models), confirming the whitening benefit is not generic perturbation regularization. *The formation benefit is therefore a probe-conditioning effect, not a representational one*: whitening flattens the covariance spectrum of the probe's own training space, which changes the optimization landscape of a 100-class linear head (direct conditioning measurements in Appendix D: whitening reduces the in-rank condition number by up to six orders of magnitude and lowers the reachable training loss at convergence, though it initially slows early optimization) — a measurement condition, not a representational improvement. The MLP-probe control (Appendix C) corroborates this: under a 2-hidden-layer MLP readout the whitening benefit vanishes (+1.1 and −3.3 pp) while the aniso harm grows (−15.2 and −16.4 pp). The preregistered cross-architecture falsifier (two ShuffleNet-V2 recipes) also upholds the conclusion — and sharpens it: in that family the probe-space arm shows no positive benefit at any dose (at most +1.0 pp), and the over-whitening dose $\gamma = -0.5$ turns clearly negative (down to −3.9 pp), so the conditioning effect is an optimization detail of the ResNet-18 feature spectrum, neither representational nor universal.

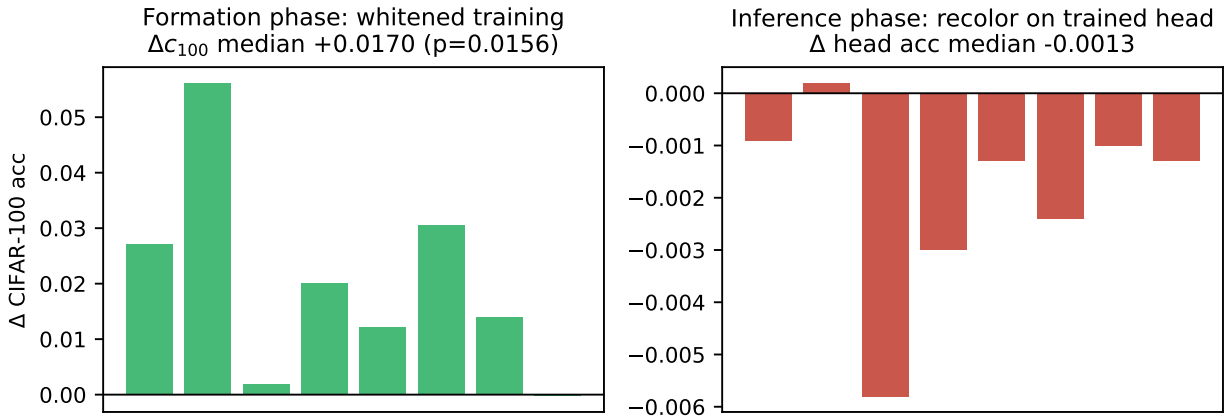


Figure 3: Phase separation of the identical whitening operator (8 paired models). *Left*: whitening the probe space slightly *improves* the measured CIFAR-100 probe (7/8 positive) — but the dose–response adjudication below shows this benefit is fully reproduced without any formation-phase manipulation: it is a probe-conditioning effect, not a representational one. *Right*: the same operator applied at inference to the unchanged trained head consistently *degrades* accuracy (7/8 negative). The effects live in the measurement, not in the geometry.

Table 2: E3 paired results (8 ResNet-18 models). c_{100}^{w}: transfer probe in the whitened space after formation-phase whitening; c_{100}^{ctrl}: same probe, control arm (original space). Δ head (inference): RECOLOR(0) applied to the unchanged trained head at inference. Auto-generated; do not edit.

Model	c_{100}^{w}	c_{100}^{ctrl}	Δc_{100}	Δ id-probe	Δ head (inf.)
`10.125_w0.0001_s0`	0.300	0.273	+2.7	+0.0	-0.1
`10.125_w0.0002_s2`	0.286	0.230	+5.6	+0.0	+0.0
`10.125_w2.5e-05_s1`	0.284	0.282	+0.2	+0.0	-0.6
`10.5_w0.0001_s1`	0.180	0.160	+2.0	+0.0	-0.3
`10.5_w0.0002_s2`	0.168	0.155	+1.2	+0.0	-0.1
`10.5_w5e-05_s2`	0.179	0.148	+3.1	+0.0	-0.2
`11.0_w2.5e-05_s0`	0.158	0.144	+1.4	+0.0	-0.1
`11.0_w5e-05_s0`	0.119	0.120	-0.0	+0.0	-0.1

Revised E3 reading. Layered on E1, the phase dependence is real but its mechanism is measurement-level: whitened spaces are *easier to probe* (a linear readout trains better there) and *harder to transplant into* an existing head trained on the original geometry (features stop matching its weight space). Neither behavior is derivable from the static value of D_{eff}, and neither constitutes acting on representation geometry Figure 3. We flag explicitly: the naive reading of the left panel — "whitening during formation improves the model" — is falsified by arm (b).

Recipe stratification (post-hoc, exploratory). Whitening benefit concentrates in sharper recipes (lr 0.125: median +2.7 pp; lr 0.5: median +2.0 pp) over the sharpest (lr 1.0: median +0.7 pp, with one model at −0.03 pp); the single negative model is the lr-1.0/wd-5e−5 recipe, whose baseline D_{eff} (0.89) is the *smallest* in the grid — consistent with a different conjecture: at this recipe the whitening operator saturates (no headroom above the floor), so measured benefit vanishes; at the same time models already sitting at high D_{eff} have little whitening headroom. *Either way this stratification was not preregistered*; we report it as exploratory (the paired-gain bootstrap CI is $[+0.2, +3.1]$ pp), and the ceiling account is a conjecture, not a tested prediction. The full paired table follows.

Table 3: E2 cross-architecture summary (4 recipes per family, 19 conditions per model). "share": inference-time RECOLOR causal slope as a percentage of the observational slope. "noise": Δc_{100} median under ANISO at matched power. All numbers auto-generated from locked data.

Architecture	obs. r	obs. slope	RECOLOR share	noise Δc_{100}	noise 4/4?
ResNet-18 (E1)	+0.97	+0.137	7.7%	−10.8 pp	8/8
ResNet-34	+0.98	+0.133	3.3%	−9.6 pp	4/4
ResNet-50	+0.98	+0.087	1.9%	−11.9 pp	4/4
ShuffleNet-V2	+0.89	+0.070	−8.3%	−13.1 pp	4/4
RegNetY-400M	+0.87	+0.114	+6.0%	−16.0 pp	4/4

6 E2: architecture transfer

We repeat the observational anchor and the two decisive separation cells (RECOLOR slope; ANISO at matched power) in four further families, each with 4 recipes (lr $\in \{0.125, 0.5\}$, wd $\in \{2\text{e}{-4}, 5\text{e}{-5}\}$, seed 0): ResNet-34, ResNet-50, ShuffleNet-V2 ×1.0, and RegNetY-400MF. MobileNetV3 was originally planned but replaced by RegNetY-400MF because its classifier head contains a non-linear bottleneck after average-pooling, so penultimate features are not the input space of the linear classifier (Appendix A).

Three regularities hold in every family (Table 3, Figure 4). *(i) The anchor survives*: $r(D_{\text{eff}}, c_{100})$ is +0.98 in both deeper ResNets and +0.87 to +0.89 in the compact families — modestly weaker, consistent with narrower recipe grids, but far from the null. *(ii) The noise channel dominates and ignores D_{eff}*: ANISO at matched power drops transfer 9.6–16.0 pp (family medians; every individual model is negative in 16/16, with per-model drops of 8.3 to 19.8 pp). The D_{eff} shift at the aniso endpoint is small for most models ($\leq 12\%$ in 10/16; up to 52% in three compact-family models whose harm is at or above their family median), so the harm does not track the geometric shift; the drop is *largest* in the compact families (−13.1 and −16.0 pp) whose observational anchors are weakest — noise harm and anchor strength dissociate. *(iii) The geometric causal share stays at the noise floor*: RECOLOR slopes span −8.3% to +6.0% of the observational slope. Permutation tests (within each model, permuting c_{100} across the dose ladder destroys any dose–response structure) place each family's noise floor at $[-6.9, 6.8]$ (r18), $[-6.9, 5.0]$ (r34), $[-2.5, 43.0]$ (r50), $[-20.2, 21.7]$ (sfn), and $[-2.8, 4.6]$% (regnet). Three of the five shares (ResNet-34, ResNet-50, ShuffleNet) lie inside their floors, so the cross-family sign wobble (ShuffleNet's is negative) is null structure, not a small real channel: a channel carrying a causal signal of the size the observational slope suggests could not flip sign under a family swap while the observational correlation stays at $r \geq 0.87$. Two shares sit marginally *above* their floors' upper bounds (ResNet-18 +7.7% vs. +6.8%; RegNetY +6.0% vs. +4.6%), and the ResNet-50 floor is wide (n=4 slope instability). For these families the null reading does not rest on the floor alone but on the sign-consistent noise harm (8/8 and 4/4 negative) and the near-zero absolute RECOLOR slopes (r18 +0.0105, r50 +0.0017, regnet +0.0068 pp per unit D_{eff}, vs. observational +0.137/+0.087/+0.114). The dissociation is not a ResNet-18 artifact; if anything, the ResNet-18 grid is the *most* generous setting for the geometric path.

7 E6: marker generality

The audit targets one marker. A natural defense of geometry-intervention methods is that D_{eff} is the wrong geometry — participation ratio (PR) or effective rank might be the manipulable one. We therefore recompute two further covariance-spectrum markers on the same ID-test subsample for all 152 E1 cells (8 models × 19 conditions) and ask whether the dissociation anatomy is D_{eff}-specific.

It is not. Both markers anchor observationally ($r(\text{PR}, c_{100}) = +0.77$, $r(\text{erank}, c_{100}) = +0.86$; both track D_{eff} at $r \geq 0.81$). Under ANISO at matched power — the cell where transfer collapses 10.8 pp — PR *rises* by a median +32% and effective rank by +29%: the noise pushes these geometric markers toward the observationally "better" regime while the model gets monotonically worse, the same wrong-signed movement the ISO cell exposes for D_{eff}, now with the harm cell itself. Under RECOLOR(0) both markers move as designed (PR +92% median, erank +144%) with the same ≤ 2 pp transfer neutrality, and ROTATE pins both exactly (relative change +0.000), re-validating the pipeline. The confounded-label reading is thus not an artifact of

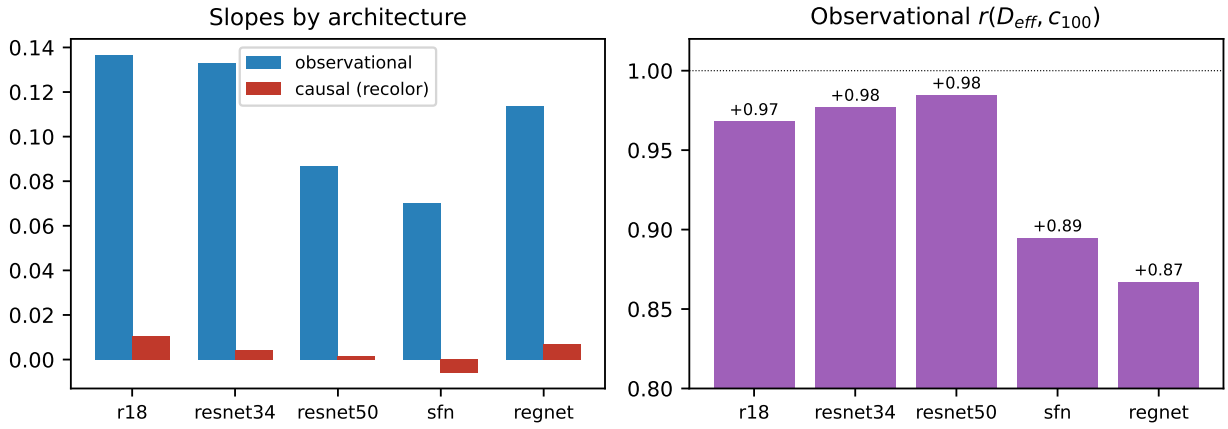


Figure 4: Cross-architecture summary (5 families, 24 recipes total). *Left*: observational vs. inference-time causal slopes per architecture — the causal bar is a small fraction of the observational bar everywhere. *Right*: the observational correlation replicates in every architecture family ($r \geq 0.87$).

one marker's definition: every spectrum-based geometry descriptor we tested is a recipe fingerprint, none is a knob.

8 Discussion: geometry as a label

The slope-decomposition arithmetic. Write the observational slope as a sum of channels: $\beta_{\text{obs}} = \beta_{\text{geo}} + \beta_{\text{recipe}} + \beta_{\text{noise}}$, where β_{geo} is the direct causal path from geometry to transfer, β_{recipe} the recipe-level confounding (the recipe sets both), and β_{noise} any channel correlated with geometry change but acting through non-geometric means. Our battery bounds the geometry-only term at a small fraction of the observational slope (point estimate $+0.0105$, 7.7% of β_{obs}; permutation floor $[-0.009, +0.009]$, with the point estimate at the floor's upper edge) and shows that the largest separable non-geometric channel (noise at matched power) moves transfer *against* the direction its D_{eff} change would suggest: $\beta_{\text{noise}} < 0$ while $\Delta D_{\text{eff}} > 0$ (a $+5.8$ median rise in D_{eff} accompanies a -6.5 pp median transfer drop). What remains of the observational association is recipe confounding. "Intervene on the geometry" methods inherit exactly this arithmetic: no detectable transmission, with a sign-reversing confounder adjacent to the knob.

What the battery does license. Three constructive conclusions survive. *(i) Noise control.* The only large causal effect we can move transfer with is noise, and it moves transfer the right way when removed: adding matched-power noise costs 6–11 pp at pinned or raised D_{eff}, while zero-noise operators of comparable geometric magnitude are transfer-neutral. Spectral-shape-matched noise is the sharpest tool: it is invisible to D_{eff} yet costs 11 pp—any noise-robustness story that monitors only geometry will miss it. *(ii) Formation-time spectral regularization.* Whitening during formation is mildly transfer-positive ($+1.7$ pp), concentrated in sharper recipes. This is a representation-shaping regularizer whose value happens to co-occur with larger D_{eff}; studying it as a geometry intervention mistakes the fingerprint for the hand. *(iii) Observational screening.* The correlation itself replicates across architectures ($r \geq 0.87$), so D_{eff} retains value as a cheap *monitoring* signal for recipe drift—it is the interventional reading that fails, not the measurement.

A checklist before reading a marker causally. The battery reduces to four questions any future "geometry intervention" result should answer: (1) *Manipulation check*: does the intervention realize the intended descriptor change ($r \geq 0.99$ here)? (2) *Operator separation*: do two operators that move the descriptor equally but differ in an orthogonal dimension (noise, timing, spectrum) move the outcome equally? (3) *Negative control*: does a descriptor-preserving operator (here ROTATE) leave the outcome unchanged? (4) *Phase separation*: does the effect survive moving the operator from formation to inference? A marker whose interventional story survives all four has earned a causal reading; D_{eff} passes none.

Relation to the companion papers. Paper 1 fixed when the observational link holds (ID-matched, architecture-adjusted, shift-typed). Paper 2 fixed what the marker measures (stability map, movement decomposition). Paper 3 (this work) fixes what it does not do: transmit causal intervention. Together they draw the full circle from measurement to mechanism.

9 Limitations

Our conclusions are scoped to: (i) *Scale and dataset*—CIFAR-10 training with CIFAR-100 transfer probes; whether ImageNet-scale recipes exhibit the same recipe-confounding structure is untested, though the mechanism (recipes jointly determine geometry and robustness) has no obvious scale dependence. (ii) *Model count*—8 ResNet-18 recipes chosen to span the widest D_{eff} range the recipe grid allows (2.4×), plus 4 recipes in each of four further architectures (ResNet-34/50, ShuffleNet-V2, RegNetY-400MF); the observational anchor and both separation results replicate in every completed family, but the grid is not exhaustively factorial. (iii) *Transfer readout*—the anchor and E1/E2 use linear probes on frozen features (the protocol that makes the observational anchor clean). The E5d control shows the headline *harm* results are not linear-probe artifacts — the aniso damage *grows* under an MLP readout (−15 to −16 pp) — but the whitening *benefit* is probe-dependent (it vanishes under MLP), and full fine-tuning could behave differently still. (iv) *One layer, one marker family*—all features are penultimate-layer, and although §7 extends the audit to participation ratio and effective rank, all three are functions of the same covariance spectrum; discrete or non-spectral descriptors (e.g. uniformity, alignment) are untested. (v) *Feature-space E3*—formation-phase whitening acts on penultimate features with a refitted head, not on weight-space training dynamics; a full-training-run whitening schedule is the natural extension, and our arm-(b) null makes the burden of proof explicit for any claim it would help representationally.

10 Conclusion

Across three preregistered causal tests, the effective dimension D_{eff} behaves consistently with a readable fingerprint of the training recipe rather than a manipulable cause of OOD generalization: spectral-shape-matched noise collapses transfer monotonically (by up to 11 pp at pinned D_{eff}); moving D_{eff} by up to 8× with zero-noise whitening never moves transfer below baseline (−0.03 to +5.6 pp); the whitening benefit in formation phase decomposes into probe conditioning, while the same operator at inference harms; the entire dissociation replicates across four further architecture families (24 recipes in total); and two further spectrum markers show the same anatomy. The observational correlation—large, stable, and real—transmits to interventions at only a small fraction of its observational size, at the upper edge of the permutation noise floor. Methods that reshape representation geometry to buy robustness should re-derive their mechanisms in these terms: what survives the battery is noise control, not geometry targeting. The broader lesson generalizes beyond D_{eff}: a strong correlation between a representation descriptor and a performance outcome is evidence about *what the recipe did*, and only a battery like this one—operator, phase, and negative controls together—can say whether the descriptor is also something the recipe can usefully be asked to change.

11 Reproducibility statement

All experiments run on public CIFAR data with fixed seeds; preregistration in `paper/adr_0016.md` (Appendices 4–5) written before unblinding; the dose–response falsification criteria (App. 8 + amendment 1) were committed before any E5 datum existed; the E5b adjudication verdict in §5 follows those preregistered branches verbatim. Scripts: `code/intervene_v2_paper.py` (E1–E3), `code/intervene_e5_dose.py` (E5 dose–response + MLP control), `code/intervene_e5c_cond.py` (E5c conditioning measurements), `code/intervene_e6_marker.py` (E6 second markers), `code/intervene_effect_probe.py`, `code/intervene_recolor.py`; data `runs/{v2_e1_full,v2_e3,v2_e2_r34,v2_e2_r50,v2_e2_sf,v2_e2_reg,v2_e5b,v2_e5d,v2_e6}.json`; analysis `code/paper3_analysis.py` + `code/paper3_tables.py` + `code/paper3_ci.py` (bootstrap CIs, permutation noise floors, Holm correction) regenerate every number, figure, and table in this paper from

the raw JSON (`runs/{...,v2_e5b,v2_e5d,v2_e5c,v2_e6}.json`; and will be released with the code and data archive).

Ethics statement

This work uses public image datasets and does not involve human subjects.

Use of AI tools

Large language models were used as writing and coding assistants. All scientific content, analyses, and reported numbers were determined and verified by the authors.

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A Preregistration and deviations

The experiment matrix and pass/fail criteria below were written and committed before any v2 number was analyzed (ADR-0016 App. 4, 2026-11-04 morning; scripts committed alongside). We reproduce the criteria verbatim in substance.

Preregistered matrix. E1: 8 ResNet-18 models $\times$ 19 conditions (base; ISO$\in \{0.5, 1, 2\}$; ANISO$\in \{0.5, 1, 2\}$; ROTATE; scale $\in \{0.5, 2\}$; RECOLOR$\in \{1.5, 0.75, 0.5, 0.25, 0\}$; total-spectrum RECOLOR$\in \{0.5, 0\}$; placebo $\in \{1, 2\}$). The ANISO(spectral-shape-matched noise) and ROTATE(spectrum-preserving rotation) cells were preregistered as the two new separation cells. E2: resnet34 / resnet50 / shufflenet_v2_x1_0 / mobilenet_v3_small $\times$ lr $\in \{0.5, 0.125\} \times$ wd $\in \{2\text{e}{-4}, 5\text{e}{-5}\} \times$ seed 0, merged with the E1 ResNet-18s for the cross-architecture anchor. E3: same whitening operator applied (a) at inference vs. (b) during formation on a 60% training subset with a refit head.

Preregistered criteria. *P1*: noise families show per-model $dc_{100}/dD_{\text{eff}} \leq -0.05$ with median ≤ -0.05 and $\geq 6/8$ same sign; zero-noise RECOLOR families |median slope| ≤ 0.02. *P1b*: ANISO at matched power with $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 0.2$ yields $\Delta c_{100} < 0$ in $\geq 6/8$ models. *P2*: cross-architecture base $r(D_{\text{eff}}, c_{100}) > 0$ in $\geq 4/5$ architectures, and inference-time RECOLOR slopes $\leq 10\%$ of the observational slope. *P3*: the formation arm lands within ± 0.01 of base c_{100} with ID head intact, while the inference arm degrades by ≥ 0.1 — phases separable.

Deviations. (1) E2 dropped mobilenet_v3_small in favor of regnet_y_400mf: the MobileNetV3 head (Howard et al., 2019) contains a non-linear bottleneck after average-pooling, so penultimate features are not the input space of the linear classifier and feature-space operators would not act on the classifier's input. RegNetY-400MF (Radosavovic et al., 2020) was chosen as a hook-clean lightweight replacement of the same efficiency-oriented family. (2) E3 protocol was revised twice *before any E3 number was analyzed*:

a first head-finetuning variant diverged on both arms (train accuracy 0.15–0.36), and a second variant omitted `model.eval()`, letting batch-norm use batch statistics and contaminate round-1 features. The final protocol (head re-initialization + per-dimension standardization + lr 10^{-2}, matching E1's verified probe) was locked before unblinding; both failed variants are archived in the experiment logs. (3) The preregistration anticipated three evaluation GPUs; two additional GPUs became available mid-run and were used only to re-shard the *same* preregistered condition matrix, not to add conditions.

Where the preregistered criteria fail as written. Full honesty requires reporting not only passes. *P1* fails as written: the ISO slope median is -0.008, far above the -0.05 threshold (its D_{eff} movement is huge, so per-unit slopes are tiny), and the ANISO slope is numerically unstable because ANISO pins D_{eff} by construction (division by a near-zero denominator). The paper's noise claim therefore rests on the P1b *endpoint*, which passes cleanly (8/8 negative, Holm $p = 0.012$), not on P1 slopes. *P2*'s share bound passes in every family. *P3* fails as written on both sub-criteria: the formation arm lands at $+0.017$, *outside* the preregistered ± 0.01 window (in the beneficial direction — this is the resurrection channel adjudicated below), and the inference harm, while directionally consistent (7/8 negative, Holm $p = 0.016$), is $30\times$ smaller than the preregistered ≥ 0.1 collapse. We interpret P3's failure as the criterion being mis-calibrated at pre-registration time, not as phase separation being absent; the mechanism adjudication below replaces it.

Amendment 1 (dose–response falsification criteria). Written and timestamped *before any E5 datum existed*, after the E3 endpoint but before the dose–response data. Three mechanism hypotheses for the formation benefit: *H-reg* (generic regularization; predicts dose-flat gains), *H-geom* (representational transmission; predicts monotone gains that persist when the whitened representation is scored in the original space), and *H-cond* (probe conditioning; predicts monotone gains fully reproduced in the probe's own transformed space, absent on original-space readouts). Adjudication rule: the paper's "geometry is not a knob" conclusion is *withdrawn* if and only if arm (b) — whitened-formation representation scored on original features — shows monotone in-dose benefits ≥ 0.5 pp in all models replicating across architectures. *Outcome*: arm (b) is flat-null in every model at every dose; arm (a) reproduces the entire benefit — H-cond confirmed, H-reg and H-geom rejected. The conclusion stands.

B E1 endpoint summary table

Rather than a 152-row dump, we release the full per-cell records as machine-readable JSON (`runs/v2/e1/intervene_v2.json`; 8 models × 19 conditions) and summarize the decisive endpoints here. Table 4 reports per-model endpoint deltas for the three decisive cells of Figure 2.

Table 4: E1 per-model endpoint deltas (pp): ANISO(2), ISO(2), and RECOLOR(0) vs. base. Negative values: transfer loss. The ANISO column is negative in 8/8 models despite $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 44\%$; the RECOLOR(0) column is never negative despite D_{eff} multiplying 1–8×. Auto-generated from the locked data.

Model	ANISO(2) Δc_{100}	ISO(2) Δc_{100}	RECOLOR(0) Δc_{100}
`l0.125_w0.0001_s0`	-13.2	-13.1	+2.7
`l0.125_w0.0002_s2`	-10.7	-10.8	+5.6
`l0.125_w2.5e-05_s1`	-13.0	-13.7	+0.2
`l0.5_w0.0001_s1`	-10.8	-5.9	+2.0
`l0.5_w0.0002_s2`	-11.1	-6.0	+1.2
`l0.5_w5e-05_s2`	-9.9	-6.6	+3.1
`l1.0_w2.5e-05_s0`	-10.5	-6.3	+1.4
`l1.0_w5e-05_s0`	-8.6	-4.8	-0.0

C E5d: MLP-probe control

To test whether the E1 findings depend on the linear readout, we repeat the decisive conditions (base, RECOLOR(0), ANISO(2), ROTATE) on two representative models with a 2-hidden-layer (512, 512, ReLU) probe under the identical optimization protocol. The *harm* results strengthen: ANISO(2) costs -15.2 and

−16.4 pp under the MLP (vs. −9.9 and −10.7 pp linear) — the noise damage is not an artifact of linear readout. ROTATE remains small (−1.6, −3.9 pp). The whitening *benefit* disappears (+1.1 and −3.3 pp vs. +3.1 and +5.6 pp linear), consistent with the probe-conditioning account of §5: whitening eases *linear* probe optimization specifically. We flag that this control covers two models only; it establishes direction, not precise magnitudes.

D E5c: direct conditioning measurements

Exploratory (not preregistered; added after the E5b adjudication to turn the conditioning inference into a measurement). For the three E5b models we train the shared-protocol 100-class linear probe in the original vs. the dose-whitened space (identical 60% training subset, seed-pinned) and record (i) conditioning statistics of the probe-train covariance and (ii) the training curve at epochs $\{1, 2, 4, 8, 16, 24, 32, 40\}$.

Because ReLU features are numerically rank-deficient, the raw $\lambda_{\max}/\lambda_{\min}$ condition number is degenerate; we report the *in-rank* condition number λ_1/λ_r instead (the numerical rank at $\lambda > 10^{-10}\lambda_{\max}$ is itself diagnostic: 494–512 of 512 in the two sharper-recipe ResNet-18 models, but only 99–122 of 512 in the lr-0.5 model, rising to 106–122 after whitening — whitening raises the usable rank there). At γ=0 whitening reduces the in-rank condition number by 10^2–$10^6\times$ (e.g. $2.8\times10^9 \to 8.4\times10^3$ in `l0.125_w0.0002_s2`), and partial doses interpolate (γ=0.75: 9.3×10^7). Two honest qualifications: first, the conditioning improvement does not track the probe-space benefit monotonically across models (the `l0.5_w5e-05_s2` model shows only a 1.9× reduction yet a +2.1 pp benefit, so the benefit is not a pure condition-number effect); second, whitening *slows early optimization* (epoch-1 training loss 4.05 → 4.50, epoch-1 accuracy 8.8% → 3.7%, consistent with the standardization in the probe protocol partially undoing the whitening early in training) while lowering the *reachable* loss at convergence (epoch-40 training loss 3.43 → 3.01 in the same model, and lower train loss at convergence in 8/9 model–dose pairs; the exception is the smallest benefit model at the $\gamma = -0.5$ dose). The probe-conditioning account of §5 therefore survives the direct measurement in a sharpened form: whitening changes where probe optimization can *end*, not how fast it starts — a property of the measurement space, not of the representation. Data: `runs/v2_e5c.json`.